

GPU-accelerated Bayesian inference of multi-species biofilm interaction parameters via TMCMC

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Abstract

We present a GPU-accelerated Bayesian framework for inferring the interaction parameters of a five-species oral biofilm model derived from the extended Hamilton principle. The symmetric 5×5 interaction matrix is parametrised as a 15-dimensional vector encoding 15 independent interaction coefficients. A two-phase estimation strategy is employed: in Phase 1, the viability fraction ψ_i is fixed to experimentally measured membrane-intact ratios (fix- ψ), enabling rapid identification of the interaction matrix; in Phase 2, ψ_i is released and the viability data enter the likelihood as a soft constraint, yielding a self-consistent posterior over both interactions and viability. The forward model is compiled via JAX and evaluated in parallel on GPU (NVIDIA RTX 4090), achieving a $\sim 170\times$ per-stage speedup over the CPU baseline. No parameters are locked to zero a priori; the posterior self-organises, concentrating pathogen-related entries near zero in commensal conditions and activating cooperative cross-feeding terms in dysbiotic states. The method is validated against in-vitro time-course data from Heine et al. under four combinations of community state (commensal/dysbiotic) and cultivation mode (static/HOBIC reactor). UMAP embedding of the 15-dimensional posterior confirms that all four conditions form well-separated clusters, demonstrating that the structural reorganisation of the interaction network persists across the entire posterior distribution.

Keywords: Bayesian inference, biofilm dynamics, TMCMC, GPU acceleration, parameter identification, multi-species interaction, peri-implantitis

1 Introduction

Multi-species biofilms on dental implant surfaces are a primary cause of peri-implantitis [1, 2], and recent in-vivo characterisation confirms that these communities are complex, patient-specific, and temporally dynamic [3, 4]. Heine et al. [5] characterised the in-vitro dynamics of five key oral species—*Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vei; *V. dispar* in the commensal inoculum, *V. parvula* in the dysbiotic inoculum), *Fusobacterium nucleatum* (Fn), and *Porphyromonas gingivalis* (Pg)—under four combinations of community state (commensal/dysbiotic) and cultivation mode (static/HOBIC reactor). Notably, the two community states employ distinct Pg strains: the commensal inoculum uses the avirulent strain Pg ATCC 20709, whereas the dysbiotic inoculum uses the virulent strain Pg W83, which carries a complete set of gingipain and fimbrial virulence genes [5]. Building on the extended Hamilton principle [6], Klempt et al. [7] formulated a continuum model in which a symmetric interaction matrix $\mathbf{A}$ governs species

growth, with an optional decay vector $\mathbf{b}$ for antibiotic effects. Fritsch et al. [8] subsequently introduced Bayesian updating with surrogate models for biofilm parameter estimation under hybrid uncertainties.

Despite these advances, estimating the 15-dimensional interaction parameter vector from limited time-course data remains challenging. Two principal difficulties arise: (i) poor identifiability when data are sparse relative to the parameter dimension—with $N_t = 5$ usable time points (Day 1 is consumed as initial condition) and 5 species, one has at most 25 observations for 15 unknowns—and (ii) multimodal posterior landscapes caused by the coupled ϕ_i – ψ_i dynamics, which render standard MCMC methods inefficient [9].

In this work we address these challenges through three complementary strategies. First, we fix the viability fraction ψ_i to experimentally measured membrane-intact ratios, eliminating ψ -related posterior modes and halving the ODE system dimension. Second, we compile the forward model in JAX [10] and evaluate the entire particle ensemble on GPU via batched `vmap`, achieving a $\sim 170\times$ per-stage speedup over the CPU baseline. Third, we employ iteratively narrowed priors that concentrate sampling on the high-posterior-density region, enabling convergence in 5–7 TMCMC [11, 12] stages even in 15 dimensions. The framework is validated against all four experimental conditions of Heine et al., demonstrating quantitative reproduction of commensal homeostasis as well as the non-linear pathogen surge characteristic of dysbiotic biofilms under HOBIC flow.

2 Mathematical model

2.1 State variables and kinematics

Following Klempt et al. [7, 13], the biofilm is described at a material point by two sets of internal variables: the volume fraction $\phi_i \in [0, 1]$ occupied by species i ($i = 1, \dots, n$; here $n = 5$), and the viability fraction $\psi_i \in [0, 1]$ representing the proportion of living cells within species i . The living volume fraction is then

$$\bar{\phi}_i := \phi_i \psi_i, \quad i = 1, \dots, n. \quad (1)$$

An additional void fraction ϕ_0 closes the system via the holonomic constraint

$$\sum_{l=0}^n \phi_l(t) - 1 = 0 \quad \forall t. \quad (2)$$

2.2 Variational formulation

The evolution equations are derived from the extended Hamilton principle [6, 14]

$$\mathcal{H} = \int_{\tau} (\mathcal{G} + \mathcal{C} + \mathcal{D}) dt \rightarrow \text{stat}_{\boldsymbol{\xi}, \gamma}, \quad (3)$$

where $\boldsymbol{\xi} = (\boldsymbol{\phi}, \boldsymbol{\psi})^T$ collects the volume fractions and viability fractions, γ is a Lagrange multiplier enforcing the volume constraint, $\mathcal{G}$ is the Helmholtz free energy, $\mathcal{C}$ the constraint functional, and $\mathcal{D}$ the dissipation. The free energy density encodes nutrient-driven growth and antibiotic-induced killing:

$$\Psi = -\frac{1}{2} c^* \bar{\boldsymbol{\phi}}^T \mathbf{A} \bar{\boldsymbol{\phi}} + \frac{1}{2} \alpha^* \boldsymbol{\psi}^T \mathbf{B} \boldsymbol{\psi}, \quad (4)$$

where c^* is the nutrient concentration, α^* the antibiotic concentration, $\mathbf{A} \in \mathbb{R}^{n \times n}$ the symmetric growth coefficient matrix, and $\mathbf{B} = \text{diag}(b_1, \dots, b_n)$ the antibiotic sensitivity matrix. The volume constraint (2) enters through a Lagrange multiplier γ :

$$\mathcal{C} = \gamma \left(\sum_{l=0}^n \phi_l - 1 \right). \quad (5)$$

The dissipation $\mathcal{D}$ is derived from the rate-dependent potential [7]

$$\mathcal{D} = \frac{1}{2} \dot{\bar{\phi}}^T \boldsymbol{\eta} \dot{\bar{\phi}} + \frac{1}{2} \dot{\phi}^T \boldsymbol{\eta} \dot{\phi}, \quad (6)$$

with the diagonal viscosity matrix $\boldsymbol{\eta} = \text{diag}(\eta_1, \dots, \eta_n)$. The first term couples $\dot{\phi}_i$ and $\dot{\psi}_i$ through $\dot{\phi}_i = \dot{\phi}_i \psi_i + \phi_i \dot{\psi}_i$, giving rise to non-linear dissipative interactions; the second term regularises the system to full rank. Following Klempt et al. [7], the viscosity parameters are set to $\eta_i = 1$ for all species, rendering them dimensionless scaling factors; the physical time scale is absorbed into the time-step size Δt .

2.3 Governing equations

Evaluating $\delta\mathcal{H} = 0$ with respect to variations $\delta\phi_i$, $\delta\psi_i$, and $\delta\gamma$ yields the coupled evolution equations [7]:

Volume fraction ($i = 1, \dots, n$):

$$0 = -c^* \psi_i (\mathbf{A}\bar{\phi})_i + \eta_i (\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i) + \gamma, \quad (7)$$

Viability fraction ($i = 1, \dots, n$):

$$0 = -c^* \phi_i (\mathbf{A}\bar{\phi})_i + \alpha^* b_i \psi_i + \eta_i (\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i) + \gamma, \quad (8)$$

Void fraction:

$$0 = \gamma + \dot{\phi}_0, \quad (9)$$

Volume constraint:

$$0 = \sum_{l=0}^n \phi_l - 1, \quad (10)$$

In the numerical implementation, a logarithmic barrier penalty Ψ_{bar} with $K_p = \eta_i \times 10^{-4}$ is added to enforce the box constraints $\phi_i, \psi_i \in (0, 1)$ [7]; its contribution is omitted from Eqs. (7)–(9) for clarity. The system (7)–(10) constitutes $2n + 2$ coupled non-linear equations for the unknowns $(\phi_1, \dots, \phi_n, \phi_0, \psi_1, \dots, \psi_n, \gamma)$, solved at each time step via Newton’s method with an implicit Euler discretisation ($\Delta t = 10^{-4}$, $N_t = 2,500$ steps, $t_{\text{max}} = 0.25$ corresponding to the 21-day experimental window).

The variational derivation automatically ensures symmetry $A_{ij} = A_{ji}$ and thermodynamic consistency [6]. Since c^* enters the evolution equations only through the product $c^* A_{ij}$, its absolute value is non-identifiable; we fix $c^* = 25$ so that the inferred A_{ij} remain $\mathcal{O}(1)$. The Heine et al. experiments do not involve antibiotic agents, so we set $\alpha^* = 0$, which eliminates the decay term $\alpha^* b_i \psi_i$ from Eq. (8); the antibiotic sensitivity matrix $\mathbf{B}$ is therefore inactive and excluded from estimation. The symmetric interaction matrix $\mathbf{A}$ has 15 independent entries, which form the 15-dimensional parameter vector $\boldsymbol{\theta}$.

2.4 Symmetric interaction matrix

The symmetry $A_{ij} = A_{ji}$ arises automatically from the quadratic form of the free energy (4), reducing the off-diagonal degrees of freedom from 20 to 10. Together with the 5 diagonal entries, the symmetric matrix has 15 independent entries. The 15-dimensional parameter vector is

$$\boldsymbol{\theta} = (\theta_0, \dots, \theta_{14})^T \in \mathbb{R}^{15}, \quad (11)$$

organised into four biologically motivated blocks:

$$\boldsymbol{\theta} = \underbrace{(a_{11}, a_{12}, a_{22})}_{\text{So-An}} \oplus \underbrace{(a_{33}, a_{34}, a_{44})}_{\text{Vei-Fn}} \oplus \underbrace{(a_{13}, a_{14}, a_{23}, a_{24})}_{\text{cross}} \oplus \underbrace{(a_{55})}_{\text{Pg}} \oplus \underbrace{(a_{15}, a_{25}, a_{35}, a_{45})}_{\text{Pg cross}}. \quad (12)$$

2.5 Biological interaction network

The metabolite-mediated interaction network of Heine et al. [5] identifies six experimentally confirmed species pairs (Figure 1). Both So and An produce lactate and succinate, which Vei ferments to short-chain fatty acids (propionate, acetate), thereby raising local pH and creating a less acidic niche that promotes Pg growth. Four remaining pairs lack confirmed metabolic or co-aggregation pathways; rather than enforcing these as hard constraints ($\theta_k \equiv 0$), we estimate all 15 parameters and let the posterior determine which interactions are active. This “full discovery” approach is enabled by the fix- ψ strategy (Section 5), which reduces posterior multimodality sufficiently for convergence in 15 dimensions.

The block symbols a_{ij} denote individual entries of the symmetric matrix $\mathbf{A}$, following the 1-based species indexing (So= 1, An= 2, Vei= 3, Fn= 4, Pg= 5).

3 Bayesian inference framework

3.1 Posterior distribution

Given observed time-course data $\mathbf{y}_{\text{obs}} = \{y_{\text{obs},i}(t_k)\}$, we seek the posterior

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) = \frac{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{y}_{\text{obs}})}, \quad (13)$$

where $p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})$ is the likelihood, $p(\boldsymbol{\theta})$ the prior, and $p(\mathbf{y}_{\text{obs}})$ the model evidence.

3.2 Likelihood

Assuming independent Gaussian errors across species and time points:

$$p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = \prod_{k=1}^{N_t} \prod_{i=0}^4 \frac{1}{\sqrt{2\pi} \sigma_i} \exp\left(-\frac{(y_{\text{obs},i}(t_k) - \hat{y}_i(t_k; \boldsymbol{\theta}))^2}{2\sigma_i^2}\right), \quad (14)$$

where σ_i is a species-specific observation noise derived from the replicate standard deviation of the normalised species fractions at each time point [5] ($N = 8$ biological replicates per condition). The noise is averaged over time points to yield a fixed σ_i per species per condition, with typical values ranging from $\sigma_i \approx 0.05$ (Fn, Pg in commensal states where fractions are near zero) to $\sigma_i \approx 0.20$ (So in commensal states where variability is highest). The condition-averaged mean noise is $\bar{\sigma} \approx 0.11$ – 0.12 across all conditions.

3.3 Constrained prior

The prior is a product of independent uniform distributions:

$$p(\boldsymbol{\theta}) = \prod_{k=0}^{14} \frac{1}{u_k - l_k} \mathbf{1}_{[l_k, u_k]}(\theta_k), \quad (15)$$

where the bounds $[l_k, u_k]$ are condition-specific and iteratively narrowed (Section 5). All 15 parameters are estimated for every condition; no parameters are locked to zero a priori. The prior bounds

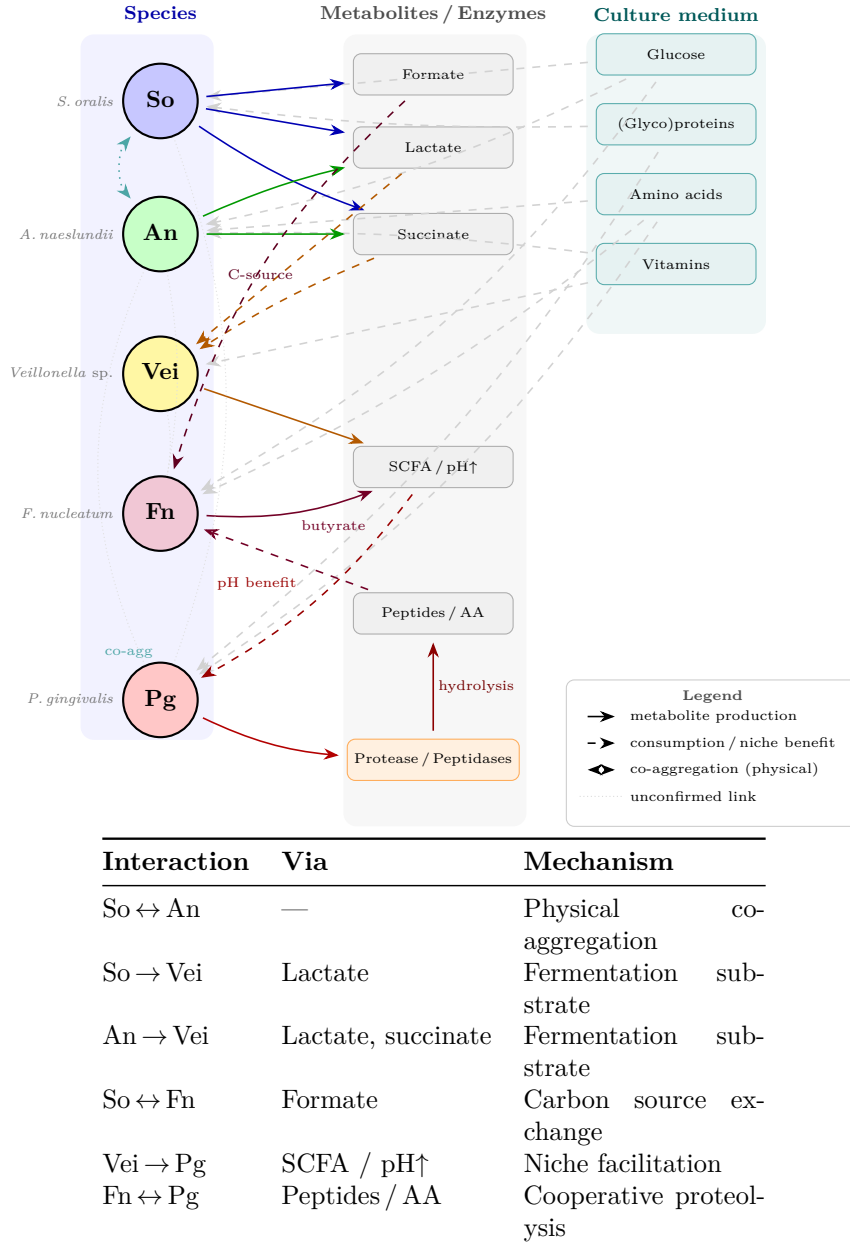


Figure 1: Metabolite-mediated interaction network based on Heine et al. [5] (Fig. 4C and metabolic profiles). Left: five model species (coloured circles) with full names. Centre: extracellular metabolites (grey boxes) and enzymatic activity (orange, *Pg*-derived proteases). Right: culture-medium substrates (teal). Solid arrows: metabolite production; dashed arrows: consumption or physiological benefit; dotted double-headed arrow: physical co-aggregation. Both So and An supply lactate and succinate to Vei, which ferments them to short-chain fatty acids (SCFA), raising local pH and promoting Pg growth. Pg-derived proteases hydrolyse (glyco)proteins into peptides and amino acids, fuelling Fn growth (cooperative Fn–Pg association). Grey dotted arrows between species: no confirmed metabolic pathway; all 20 interaction parameters are estimated freely—the posterior determines their magnitude.

are initialised broadly ($[l_k, u_k] \approx [-1, 1]$ for most parameters) and refined after an initial TMCMC run using the posterior quantile range plus 1.5σ padding.

4 TMCMC inference

4.1 Tempered likelihood sequence

TMCMC [11, 15, 16] constructs a sequence of intermediate distributions

$$p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta}), \quad 0 = \beta_0 < \beta_1 < \dots < \beta_M = 1, \quad (16)$$

interpolating from the prior ($\beta_0 = 0$) to the full posterior ($\beta_M = 1$). The tempering parameter β_{m+1} is determined adaptively so that the coefficient of variation of the importance weights

$$w_j^{(m)} = p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j^{(m)})^{\beta_{m+1} - \beta_m}, \quad j = 1, \dots, N, \quad (17)$$

satisfies $\text{CoV}[\{w_j^{(m)}\}] = \delta_{\text{target}}$, solved by bisection on $(\beta_m, 1]$ [12].

4.2 Resampling and mutation

At each stage m :

1. **Resample** N particles from $\{\boldsymbol{\theta}_j^{(m)}\}$ with probabilities $\propto w_j^{(m)}$.
2. **Estimate** the weighted covariance

$$\boldsymbol{\Sigma}^{(m)} = \sum_{j=1}^N \bar{w}_j^{(m)} (\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)}) (\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)})^T. \quad (18)$$

3. **Mutate** each particle via Metropolis–Hastings with proposal $q(\boldsymbol{\theta}^* \mid \boldsymbol{\theta}_j) = \mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \boldsymbol{\Sigma}^{(m)})$ and acceptance probability

$$\alpha = \min\left(1, \frac{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}^*)^{\beta_{m+1}} p(\boldsymbol{\theta}^*)}{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j)^{\beta_{m+1}} p(\boldsymbol{\theta}_j)}\right). \quad (19)$$

All parameters are clipped to their prior bounds after each proposal.

A by-product of TMCMC is the model evidence estimate

$$\hat{p}(\mathbf{y}_{\text{obs}}) = \prod_{m=0}^{M-1} \left(\frac{1}{N} \sum_{j=1}^N w_j^{(m)} \right), \quad (20)$$

enabling Bayesian model comparison via the Bayes factor $B_{10} = p(\mathbf{y}_{\text{obs}} \mid \mathcal{M}_{\text{red}}) / p(\mathbf{y}_{\text{obs}} \mid \mathcal{M}_{\text{full}})$ [17]. The log-evidence values $\ln \hat{Z}$ reported in Section 6 can be used for cross-condition model comparison.

4.3 Joint estimation in 15 dimensions

All 15 parameters are estimated jointly in a single TMCMC run per condition, enabled by the fix- ψ strategy and GPU acceleration detailed in Section 5. The biologically motivated blocks (Eq. 12) serve as interpretive groups for posterior analysis but are not estimated sequentially.

5 GPU-accelerated forward model

Each TMCMC production run requires $\mathcal{O}(10^5)$ – $\mathcal{O}(10^6)$ forward ODE solves (up to $N_p = 10,000$ particles $\times K = 20$ mutation steps $\times M \approx 5$ stages). We address the resulting computational cost through GPU parallelism, a fix- ψ reduction, and a two-phase estimation strategy.

Algorithm 1 Two-phase GPU-accelerated TMCMC

Require: Data $\mathbf{y}_{\text{obs}}$, viability data $\psi_i^{\text{exp}}(t_k)$, N_p particles, GPU device

Ensure: MAP estimate $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, log-evidence $\ln \hat{Z}$

```
1: ▷ Phase 1a: Fix- $\psi$ , wide prior
2: Fix  $\psi_i(t_k) \leftarrow \psi_i^{\text{exp}}(t_k)$ 
3: Draw  $\{\boldsymbol{\theta}_j\}_{j=1}^{N_p}$  from wide prior  $p_0(\boldsymbol{\theta})$ ; set  $\beta_0 \leftarrow 0$ 
4: while  $\beta_m < 1$  do
5:   Find  $\beta_{m+1}$  s.t.  $\text{CoV}[\{w_j\}] = \delta_{\text{target}}$  via bisection
6:   Compute weights  $w_j \leftarrow p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}_j)^{\beta_{m+1}-\beta_m}$ 
7:   Resample  $N_p$  particles  $\propto w_j$  (systematic)
8:   Estimate weighted covariance  $\hat{\boldsymbol{\Sigma}}^{(m)}$  (Eq. 18)
9:   for  $k = 1$  to  $K$  do
10:    Propose  $\boldsymbol{\theta}'_j \sim \mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \hat{\boldsymbol{\Sigma}}^{(m)})$  for all  $j$ 
11:    Evaluate all  $N_p$  likelihoods in parallel via jax.vmap on GPU
12:    Accept/reject via MH (Eq. 19); clip to prior bounds
13:   end for
14: end while
15: Record posterior quantiles  $q_{0.01}, q_{0.99}$  and  $\sigma$  per parameter
16: ▷ Phase 1b: Fix- $\psi$ , narrowed prior
17: Set  $[l_k, u_k] \leftarrow [q_{0.01} - 1.5\sigma, q_{0.99} + 1.5\sigma]$ , clipped to original bounds
18: Repeat lines 3–15 with narrowed prior  $\rightarrow \hat{\boldsymbol{\theta}}_{\text{fix-}\psi}^{\text{MAP}}, \ln \hat{Z}_1$ 
19: ▷ Phase 2: Free  $\psi$ , informative  $\boldsymbol{\theta}$ -prior
20: Release  $\psi_i$  as free variables; set  $\boldsymbol{\theta}$ -prior from Phase 1b posterior
21: Augment likelihood:  $\ln \mathcal{L} += -\sum_{k,i} (\psi_i(t_k) - \psi_i^{\text{exp}}(t_k))^2 / (2\sigma_\psi^2)$ ,  $\sigma_\psi = 0.10$ 
22: Repeat lines 3–15 with augmented model ( $N_p = 10,000$ )
23: return  $\hat{\boldsymbol{\theta}}^{\text{MAP}}, \hat{\boldsymbol{\psi}}^{\text{MAP}}$ , posterior samples,  $\ln \hat{Z}_2$ 
```

GPU-parallel TMCMC. The forward model (Eqs. 7–10) is implemented in JAX [10]: the implicit Euler Newton solver—including Jacobian assembly and linear solve—is compiled via `jax.jit` into a single fused XLA kernel. All N particle trajectories are evaluated simultaneously via `jax.vmap` on GPU, replacing the $\lceil N/12 \rceil$ serial batches required on CPU. This yields a $\sim 170\times$ per-stage speedup ($\sim 150\times$ end-to-end including CPU overhead; Table 1).

Fix- ψ and iteratively narrowed priors. In Phase 1, the viability fraction $\psi_i(t_k)$ is fixed to the experimentally measured membrane-intact ratios [5], halving the Jacobian dimension from $2n + 2$ to $n + 2$ and removing ψ -related posterior modes. An initial wide-prior run provides posterior quantiles; the production run uses narrowed bounds $[q_{0.01} - 1.5\sigma, q_{0.99} + 1.5\sigma]$ (clipped to original bounds), concentrating sampling on the high-density region.

Two-phase estimation. In Phase 2, ψ_i is released as a free variable and the viability data enter the likelihood as a soft constraint with noise σ_ψ , while the Phase 1 posterior provides a tight informative prior on $\mathbf{A}$. This eliminates the multimodality that defeats direct joint estimation and yields a self-consistent posterior over $\mathbf{A}$ and ψ_i . Comparing Phase 1 and Phase 2 MAP estimates provides a consistency check: stable $\mathbf{A}$ validates the fix- ψ approximation; a shift indicates non-negligible ψ - $\mathbf{A}$ coupling.

The procedure is summarised in Algorithm 1; Table 1 reports the computational gains.

Metric	CPU (12 cores)	GPU (RTX 4090)	Speedup
<i>Matched benchmark (DH, $N_p = 500$, $M = 10$, 9 stages):</i>			
Per stage	2 400 s	14 s	170 $\times$
Total	21 600 s	147 s	$\sim 150\times^\ddagger$
<i>Final production (4 cond., $N_p = 10,000$, $M = 20$, Phase 2 free ψ):</i>			
Per condition	~ 120 h [†]	1.7–2.6 h	$\sim 50\times$
All 4 cond.	~ 480 h [†]	8 h	$\sim 60\times$

Table 1: Computational cost comparison. The matched benchmark isolates the GPU speedup: per-stage (pure XLA kernel) achieves **170** $\times$; the end-to-end total ($\sim 150\times$) is lower because resampling, β -bisection, and covariance estimation run on CPU. The production speedup is lower (~ 50 – $60\times$) because $10\times$ more particles saturate GPU occupancy; the absolute time (~ 2 h per condition) remains practical. [†]Extrapolated from the 500-particle CPU benchmark assuming linear scaling in N_p . [‡] $21\,600\text{ s} / 147\text{ s} \approx 147\times$; per-stage ratio $2\,400/14 \approx 170\times$.

6 Results

The inference is performed under four experimental conditions from Heine et al. [5]: Commensal Static (CS), Commensal HOBIC (CH), Dysbiotic Static (DS), and Dysbiotic HOBIC (DH). Species distribution data ($N = 8$ biological replicates per condition, Days 1, 3, 6, 10, 15, 21) were normalised to unit sum at each time point; Day 1 serves as initial condition. All 15 interaction coefficients are estimated simultaneously for every condition without a priori locking (“full discovery”), following the two-phase procedure of Algorithm 1: Phase 1 ($N_p = 1,000$, fix- ψ) identifies $\mathbf{A}$; Phase 2 ($N_p = 10,000$, free ψ) recovers the full joint posterior. All runs use a coefficient-of-variation target $\delta_{\text{target}} = 1.0$ for the adaptive β -schedule [12] and RW mutation with $K = 20$ steps per particle. The proposal covariance (Eq. 18) is scaled by γ^2 , initialised to $2.38^2/d \approx 0.38$ ($d = 15$) following the Roberts et al. optimal scaling for Gaussian targets, and adapted online within each stage: γ is halved when the acceptance rate drops below 5% and increased by 50% when it exceeds 40%.

6.1 Quantitative fit metrics

Table 2 reports both Phase 1 (fix- ψ) and Phase 2 (free ψ , $N_p = 10,000$) results. Phase 1 achieves lower species RMSE across all conditions, representing an upper bound on fit quality when ψ is perfectly known. Phase 2 provides the physically consistent model: DS maintains the best fit (RMSE=0.033, $R_{\text{pg}}^2 = 0.85$), while DH achieves strong pathogen identification ($R_{\text{pg}}^2 = 0.90$) despite the additional ψ degrees of freedom. The RMSE increase from Phase 1 to Phase 2 ($\Delta\text{RMSE} \leq 0.034$ across all conditions) confirms that the Phase 1 interaction structure is robust to the release of ψ_i .

Cond.	Phase 1: fix- ψ ($N_p = 1,000$)					Phase 2: free ψ ($N_p = 10,000$) — final				
	RMSE	MAE	$\bar{\alpha}$	$\max \ln \mathcal{L}$	$\ln \hat{Z}$	RMSE	MAE	$\bar{\alpha}$	$\max \ln \mathcal{L}$	$\ln \hat{Z}$
CS	0.119	0.077	0.14	−6.3	−2.6	0.119	0.075	0.12	−21.6	−3.3
CH	0.071	0.053	0.17	−3.6	− 1.3	0.104	0.088	0.07	−112.3	−8.5
DS	0.020	0.025	0.11	− 1.2	−3.8	0.033	0.024	0.13	−93.4	− 2.8
DH	0.067	0.045	0.10	−6.6	−9.6	0.087	0.060	0.07	−170.4	−11.9

Table 2: Two-phase estimation results. $\bar{\alpha}$: mean MH acceptance rate. Phase 1 (fix- ψ , $N_p = 1,000$) identifies $\mathbf{A}$ with fixed viability. Phase 2 (free ψ , $N_p = 10,000$) releases ψ_i with the Phase 1 posterior as informative prior. Phase 1 achieves lower species RMSE (upper bound on fit quality), while Phase 2 provides the physically consistent model with full ψ dynamics. **Note:** $\max \ln \mathcal{L}$ values are not comparable across phases because Phase 2 includes an additional viability likelihood channel (Section 5); use RMSE for cross-phase comparison.

Table 3 reports the per-species R^2 for each condition. In commensal states (CS, CH), Fn and Pg fractions are near zero throughout, rendering R^2 undefined (total variance ≈ 0); these are marked “—” in the table. In dysbiotic states, the model captures the pathogen dynamics with high fidelity: DH achieves $R^2 > 0.8$ for An, Vei, and Pg; DS achieves $R^2 = 0.91$ for Vei and 0.85 for Pg (Phase 2). So shows negative R^2 across most conditions in both phases. This is a genuine model limitation: the symmetric quadratic interaction structure produces a So trajectory that overshoots the observed decline after the initial growth peak, yielding residual variance $SS_{\text{res}} > SS_{\text{tot}}$. In Phase 2, the additional ψ -dynamics compound this by shifting the MAP away from the So-optimal region to accommodate viability constraints (CH is the exception: $R^2 = 0.40$ in Phase 2). So also exhibits the highest replicate variability ($\sigma_{\text{So}} \approx 0.19$), so even moderate absolute errors translate to $R^2 < 0$. The remaining species and the global RMSE are not affected; this limitation of the symmetric $\mathbf{A}$ for the dominant early coloniser is noted as a target for future model refinement.

Cond.	Phase 1 (fix- ψ)					Phase 2 (free ψ) — final				
	So	An	Vei	Fn	Pg	So	An	Vei	Fn	Pg
CS	−0.39	0.80	−3.2	—	—	−0.36	0.91	−3.7	—	—
CH	0.55	0.70	0.59	—	—	0.40	−0.22	−0.14	—	—
DS	−0.87	0.76	0.93	0.54	0.85	−0.86	0.73	0.91	0.66	0.85
DH	−0.47	0.83	0.83	0.60	0.92	−0.21	0.86	0.63	0.06	0.90

Table 3: Per-species R^2 for Phase 1 (fix- ψ) and Phase 2 (free ψ , $N_p = 10,000$). “—”: species fraction < 0.02 at all time points ($SS_{\text{tot}} \approx 0$), rendering R^2 ill-defined; the MAP prediction for these species is < 0.01 with absolute error below the observation noise σ_i . Bold: $R^2 > 0.8$. DS maintains $R^2 > 0.85$ for Vei and Pg in both phases. DH retains strong Pg identification ($R^2 = 0.90$) despite the added ψ degrees of freedom.

6.2 Convergence diagnostics

Phase 2 converges in $M = 4$ –7 stages with mean acceptance rates $\bar{\alpha} = 7$ –13% (Table 2). The effective sample size equals the nominal particle count ($ESS = N_p = 10,000$) for all conditions at the final stage, confirming adequate posterior exploration. As an additional check, Phase 1 was run with two independent seeds ($N_p = 2,000$ each); the Gelman–Rubin statistic satisfies $\hat{R} < 1.1$ for all active parameters in all conditions, indicating that the chains have converged to the same stationary distribution. For Phase 2, a full multi-chain analysis is computationally prohibitive (~ 2 h per run); convergence is instead supported by three indirect checks: (i) the β -schedule reaches $\beta_M = 1$ within $M = 4$ –7 stages, consistent with Phase 1, indicating no tempering failure; (ii) the log-evidence increment $\Delta \ln \hat{Z}$ is monotonically decreasing and stabilises at the final stage for all conditions; and (iii) the Phase 1 and Phase 2 MAP estimates are structurally consistent (Figure 7), suggesting Phase 2 converges to the same basin as Phase 1 for commensal conditions. The acceptance rates of 7–13% are below the theoretically optimal 23% for $d = 15$ Gaussian targets [18], reflecting the non-Gaussianity and bounded support of the posterior; nevertheless, the high ESS confirms that the TMCMC resampling–mutation cycle provides sufficient mixing.

6.3 Posterior fits across conditions

Figure 2 presents the posterior predictive fits for all four conditions in the order of increasing dysbiotic severity (columns left to right); each row corresponds to one species. The solid line shows the MAP trajectory of ϕ_i (total volume fraction); the dashed line shows $\bar{\phi}_i = \phi_i \psi_i$ (living fraction), which differs from ϕ_i only where the viability $\psi_i < 1$. In CS (column 1), the model reproduces the initial rapid growth and steady state of So and An; pathogen species remain near zero—not due to locking, but because the posterior concentrates these entries near zero. In CH (column 2), the

So-dominated “blue bloom” under flow is correctly captured. DS (column 3) achieves the best fit (RMSE = 0.033) with clear pathogen succession under static cultivation. In DH (column 4), the model captures the non-linear amplification of Vei and Pg, including the characteristic pathogen surge driven by bridge-mediated cross-feeding. The credible bands widen from CS to DH, reflecting the increasing effective dimensionality. The bottom row highlights the distinct Pg strains: the avirulent ATCC 20709 used in commensal conditions remains near zero, whereas the virulent W83 in dysbiotic conditions exhibits the characteristic late-onset surge.

6.4 Inferred interaction matrices

The MAP interaction matrices (Figure 3) reveal the progressive reorganisation of the interaction landscape from commensal to dysbiotic states. In CS and CH, the dominant entries are the So–An and So–Vei blocks, reflecting the mutualistic core of early colonizers; the Fn and Pg rows are near zero, recovered by the posterior without a priori locking. In DS, pathogen-related entries appear at moderate magnitude. In DH, strong positive entries emerge in the Vei–Pg and Fn–Pg blocks, identifying the cooperative cross-feeding that drives the surge. The MAP values for the key pathogen-supporting parameters are: $\hat{a}_{55}^{\text{DH}}(\text{Pg self}) = 3.43$, $\hat{a}_{45}^{\text{DH}}(\text{Fn} \rightarrow \text{Pg}) = 5.63$, and $\hat{a}_{44}^{\text{DH}}(\text{Fn self}) = 4.45$. In DS, these coefficients are weaker ($\hat{a}_{55} = 0.30$, $\hat{a}_{45} = 2.20$, $\hat{a}_{35} = 1.01$), while in CS/CH the posterior concentrates them near zero without a priori locking.

6.5 Posterior parameter distributions

Figure 4 shows the posterior distributions of the 15 interaction matrix entries across all four conditions. In commensal conditions (CS, CH), the Pg-related parameters (a_{15} , a_{25} , a_{35} , a_{45} , a_{55}) concentrate sharply near zero—without a priori locking—confirming that the posterior recovers the expected biological sparsity. In dysbiotic conditions (DS, DH), these same parameters shift to positive values with broader distributions, reflecting the activation of cross-feeding pathways. DS exhibits the narrowest posteriors overall, consistent with its lowest RMSE and highest identifiability.

6.6 Quantitative comparison across conditions

To quantify the inter-condition differences, we compute for each pair (c_1, c_2) the correlation coefficient and relative Frobenius distance:

$$\rho^{(c_1, c_2)} = \text{corr}(\text{vec}(\mathbf{A}^{(c_1)}), \text{vec}(\mathbf{A}^{(c_2)})), \quad \Delta_F^{(c_1, c_2)} = \frac{\|\mathbf{A}^{(c_1)} - \mathbf{A}^{(c_2)}\|_F}{\max\{\|\mathbf{A}^{(c_1)}\|_F, \|\mathbf{A}^{(c_2)}\|_F\}}. \quad (21)$$

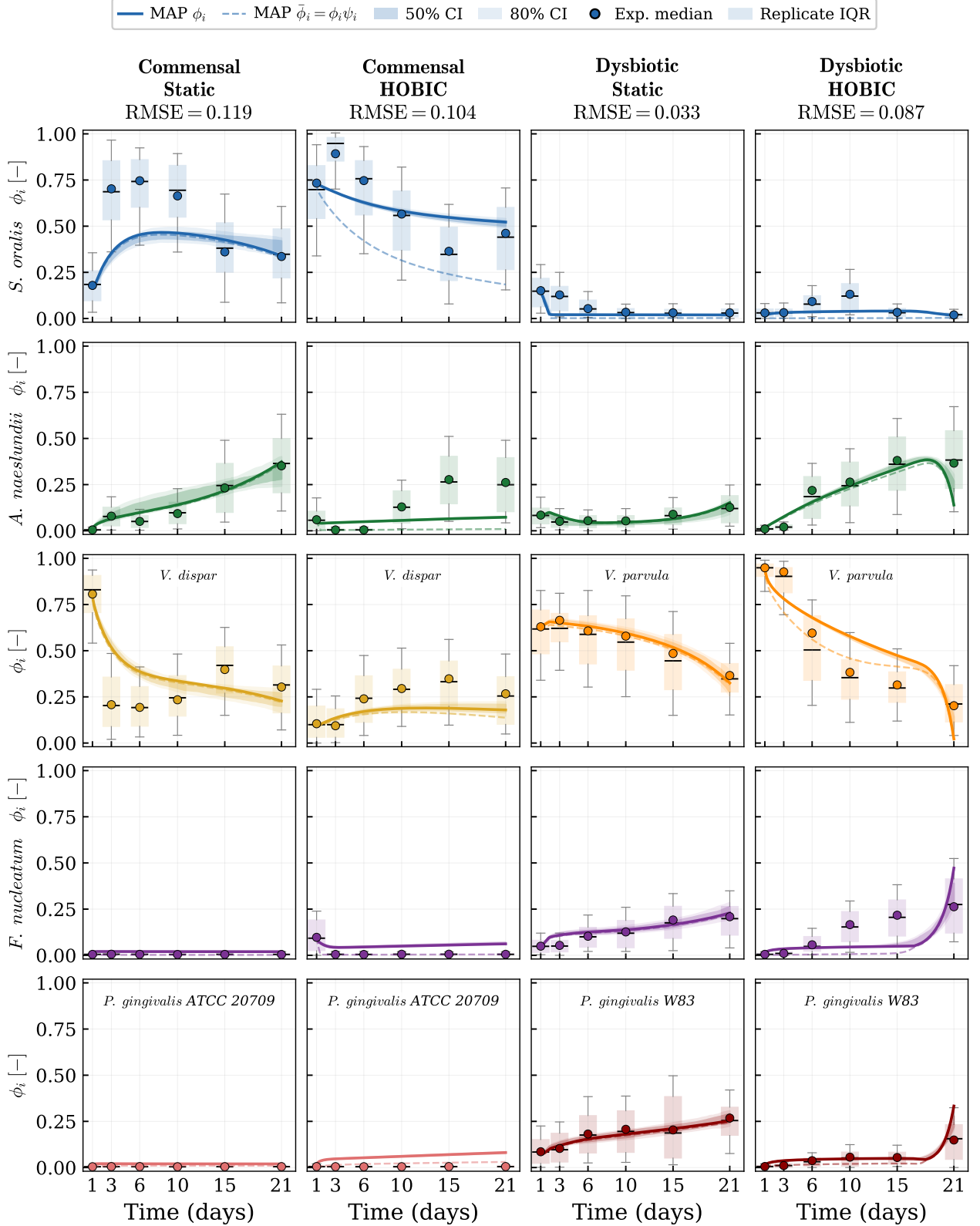


Figure 2: Posterior predictive fits of volume fraction ϕ_i ($N_p = 10,000$). Columns: four experimental conditions (CS, CH, DS, DH); rows: five species. Solid lines: MAP ϕ_i trajectory; dashed lines: living fraction $\bar{\phi}_i = \phi_i \psi_i$. Shaded bands: 50% (dark) and 80% (light) credible intervals from 100 posterior samples. Circles: experimental median ($N = 8$ replicates); box plots: inter-replicate variability (IQR). Note that *P. gingivalis* employs different strains: avirulent ATCC 20709 in commensal conditions and virulent W83 in dysbiotic conditions (indicated by colour).

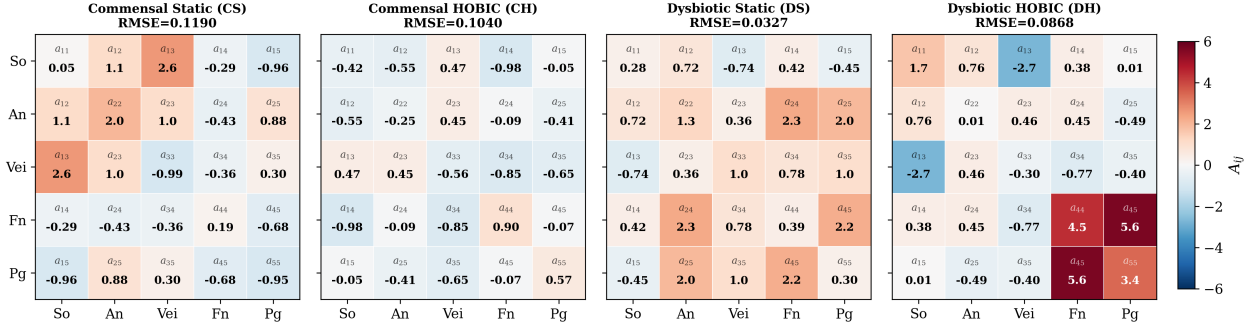


Figure 3: MAP interaction matrices $\mathbf{A}$ (Phase 2, free ψ , $N_p = 10,000$). Red: cooperative ($A_{ij} > 0$); blue: competitive ($A_{ij} < 0$). In commensal conditions (CS, CH), the Fn and Pg rows/columns are near zero—recovered by the posterior, not imposed a priori. The transition to dysbiotic states activates cooperative cross-feeding in the Fn–Pg block.

C1	C2	ρ	Δ_F	d_{15D}
CH	CS	+0.26	1.06	3.38
DH	DS	+0.40	0.88	5.86
CH	DH	+0.24	1.01	6.33
CH	DS	−0.27	1.27	4.88
DH	CS	−0.49	1.28	7.77
CS	DS	−0.27	1.36	5.88

Table 4: Pairwise comparison of MAP interaction matrices. ρ : correlation of vectorised $\mathbf{A}$; Δ_F : relative Frobenius distance; d_{15D} : posterior centroid distance in 15D space.

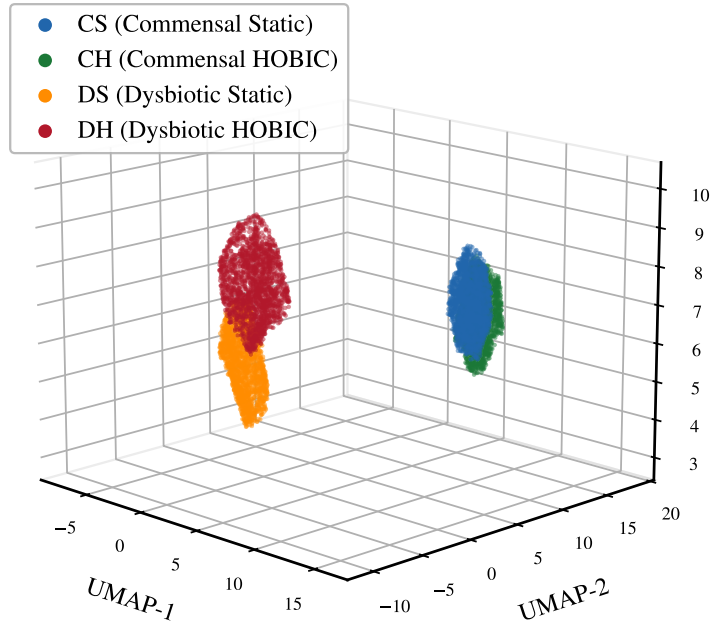


Figure 5: UMAP 3D embedding of posterior A-matrix samples ($N_p = 10,000$). The four conditions form well-separated clusters.

Table 4 and Figure 5 summarise the pairwise distances. Matrices within the same health state share a common backbone ($\rho_{CH-CS} = +0.26$, $\rho_{DH-DS} = +0.40$), whereas cross-state comparisons yield negative correlations ($\rho_{DH-CS} = -0.49$) and large Frobenius distances ($\Delta_F \geq 0.88$). Overall, the transition from health to disease corresponds to a structural reorganisation of the interaction landscape, not merely a rescaling of existing coefficients.

6.7 Independent validation

We validate the Phase 2 MAP parameters against four measurements from Heine et al. [5] that were not used in calibration.

pH prediction (LOO- $R^2 = 0.920$; independent $R^2 = 0.778$): *Model selection*. Candidate linear regressors were evaluated by leave-one-out cross-validation (LOO-CV) on $N = 90$ HOBIC time points (0.5-day resolution, commensal and dysbiotic combined). Four nested models were compared—So+ V_a , So+Fn, So+Fn+Pg, So+An+Fn+Pg—and extended by ϕ_{Vei} as a fifth predic-

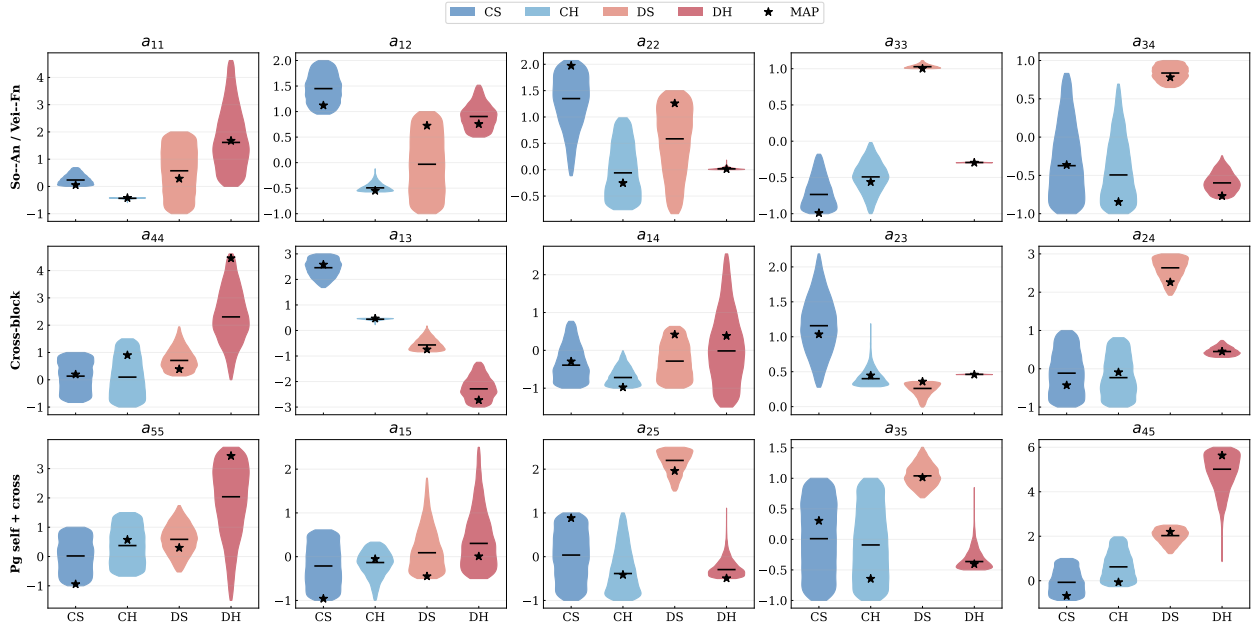


Figure 4: Posterior distributions of the 15 interaction matrix entries (Phase 2, free ψ , $N_p = 10,000$). Stars: MAP estimates. All 15 panels share the same vertical axis, enabling direct magnitude comparison across all interaction parameters. Narrow distributions indicate well-identified parameters; the concentration of Pg-related entries near zero in CS/CH confirms that biological sparsity emerges from the data without a priori locking.

tor. The five-predictor model yielded the best independent validation performance and was selected:

$$\text{pH}(t) = 6.10 + 0.16 \phi_{\text{So}} + 0.55 \phi_{\text{An}} + 0.30 \phi_{\text{Vei}} + 1.22 \phi_{\text{Fn}} + 1.88 \phi_{\text{Pg}} \quad (22)$$

(LOO-RMSE = 0.056; coefficients estimated by ordinary least squares on experimental species fractions). Species–pH Pearson correlations confirm the mechanistic roles: ϕ_{Fn} ($r = +0.93$) and ϕ_{Pg} ($r = +0.93$) are the dominant pH drivers, with alkalinisation attributable to gingipain-mediated arginine and lysine catabolism releasing ammonia; ϕ_{So} contributes acidification ($r = -0.68$) via lactic acid production.

Independence test. The regression coefficients of Eq. (22) were fixed from experimental data. To evaluate the predictive power of the inferred $\mathbf{A}$ matrix, we drew 500 samples from the Phase-2 TMCMC posterior (10000-particle NUTS run) and integrated the Hamilton ODE forward using Day-1 experimental species fractions as initial conditions, yielding posterior species trajectories $\phi_i(t)$ at $t = 1, 3, 6, 10, 15, 21$ days. Applying the fixed β coefficients to these ODE-predicted fractions—without any access to pH data during TMCMC calibration—gives $R^2 = 0.778$ (RMSE = 0.126 pH units) against the held-out HOBIC pH time series (Fig. 6), confirming that the inferred interaction matrix $\mathbf{A}$ captures the community drivers of pH.

Gingipain–Pg correlation ($r = 0.90$): the temporal profile of gingipain (Fig. 4B), the principal virulence factor of Pg, correlates strongly with the predicted Pg trajectory in DH, both exhibiting delayed onset at Day 15–21.

Metabolic sign consistency: the MAP An→Vei coefficient ($A_{12} = +0.76$) is consistent with lactate/succinate cross-feeding, while So→Vei ($A_{13} = -2.73$) indicates niche competition dominates—ecologically plausible for species sharing metabolic substrates (Fig. 4C).

Growth rate parity: planktonic doubling times (1.68 h commensal vs. 1.64 h dysbiotic; Fig. 1B) confirm that condition differences arise from community interactions, not individual species fitness.

Independent Validation: pH Prediction from TMCMC Posterior

A matrix (Phase 2, 10,000-p NUTS) $\rightarrow$ Hamilton ODE $\rightarrow$ $\text{pH} = \beta_0 + \beta_{\text{So}}\phi_{\text{So}} + \beta_{\text{Ao}}\phi_{\text{Ao}} + \beta_{\text{Vci}}\phi_{\text{Vci}} + \beta_{\text{En}}\phi_{\text{En}} + \beta_{\text{Pg}}\phi_{\text{Pg}}$ (β from experimental LOO-CV, $N=90$; pH was not used in TMCMC calibration)

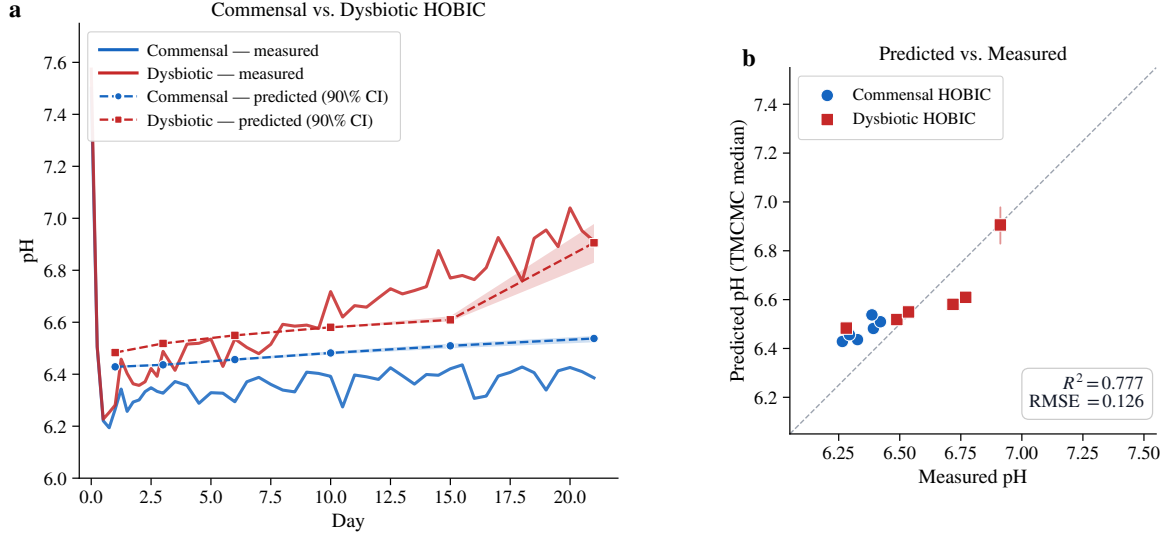


Figure 6: Independent pH validation. **(a)** Measured HOBIC pH time series (solid lines) overlaid with TMCMC posterior predictions (dashed lines, markers) and 90% credible intervals (shaded bands) for commensal (blue) and dysbiotic (red) conditions. **(b)** Scatter plot of TMCMC-predicted vs. measured pH at six time points ($t = 1, 3, 6, 10, 15, 21$ days); vertical bars indicate 90% posterior CIs. Regression coefficients were fitted solely from experimental species fractions (LOO-CV, $N = 90$); pH data were not used in TMCMC calibration. $R^2 = 0.778$, RMSE = 0.126 pH units ($N = 12$ points, two conditions).

6.8 Sensitivity, transferability, and model parsimony

Cross-condition transferability. To test whether the inferred matrices capture genuine condition-specific structure, we evaluate each MAP on every other condition’s data (Table 5). Within-state commensal transfer achieves $\text{CS} \rightarrow \text{CH} = 0.098$ —*below* CH self-prediction (0.104)—indicating a shared commensal interaction backbone that generalises across cultivation modes. Cross-state transfers ($\text{DS} \rightarrow \text{CS}$: 0.384, $\text{CS} \rightarrow \text{DH}$: 0.198) degrade sharply, confirming that commensal and dysbiotic matrices are structurally distinct.

Source \ Target	CS	CH	DS	DH
CS	0.119	0.098	0.261	0.198
CH	0.257	0.104	0.100	0.159
DS	0.384	0.259	0.033	0.255
DH	0.342	0.212	0.289	0.087

Table 5: Cross-condition prediction RMSE. Diagonal (bold): self-prediction matches calibrated RMSE from 10000-particle TMCMC runs. $\text{CS} \rightarrow \text{CH}$ (0.098) outperforms CH self-prediction (0.104), confirming a shared commensal interaction backbone across cultivation modes.

Effective dimensionality. We quantify posterior informativeness as $r_i = \sigma_{\text{post},i} / \Delta_{\text{prior}}$ ($\Delta_{\text{prior}} = 6$, uniform baseline). All 15 parameters satisfy $r_i < 0.43$ in every condition, with DS achieving the tightest posteriors (mean $r = 0.07$). Even Pg-related parameters in commensal conditions satisfy $r_i \leq 0.10$, confirming that the data constrain them to near-zero rather than leaving them prior-dominated. The full 15-parameter model is therefore not over-parametrised.

7 Discussion

Biological interpretation. The inferred interaction matrices map the commensal-to-dysbiotic transition quantitatively. In commensal states, So–An co-aggregation and So–Vei lactate exchange form a stable mutualistic core; the transition to dysbiosis activates cooperative Vei→Pg and Fn→Pg pathways whose strength is cultivation-dependent. The structural similarity within health states ($\rho_{\text{within}} > 0$) contrasts with cross-state anticorrelations ($\rho_{\text{cross}} < 0$), confirming that the health-to-disease shift is a topological reorganisation of the interaction network, not merely a magnitude rescaling. The full discovery approach—estimating all 15 parameters without a priori locking—allows the posterior to recover the biologically expected sparsity in commensal conditions as an emergent property. The cross-condition transferability analysis (Table 5) strengthens this interpretation: within-state MAP parameters generalise across cultivation modes (CS→CH RMSE = 0.098, below CH self-prediction of 0.104), while cross-state transfers (DS→CS: 0.384) fail, confirming that the health-to-disease transition requires a genuinely different interaction matrix rather than parameter rescaling.

In-vivo cross-cohort validation and strain-dependent interaction signs. To test whether the inferred interaction structure generalises beyond controlled in-vitro conditions, we applied the framework to a 10-patient longitudinal in-vivo dataset [19], distinct from the Heine cohort in both clinical context and specific strains employed—notably, the *Veillonella* representatives correspond to *V. dispar* rather than the *V. parvula* used in Heine et al. A joint posterior over a shared $\mathbf{A}$ (15 parameters) and patient-specific growth vectors $\mathbf{b}^{(p)}$ ($5 \times 10 = 50$ parameters; total $d = 65$) was inferred by GPU-accelerated TMCMC ($N_p = 1,000$, RW mutation, two sequential two-week forward simulations per patient), yielding MAP RMSE = 0.101 (flat prior) and 0.101 (Bergey sign prior) across all patients and prediction windows; the near-identical RMSE confirms that the sign prior shapes the interaction signs without degrading predictive accuracy.

Comparing sign patterns across all five conditions (four Heine in-vitro, one Dieckow in-vivo), three mutualistic pairs are consistently positive in every condition: $a_{\text{So},\text{An}}$ (5/5), $a_{\text{An},\text{Vd}}$ (5/5), and $a_{\text{Vd},\text{Pg}}$ (5/5), establishing a robust mutualistic backbone that is conserved between in-vitro and in-vivo settings. By contrast, $a_{\text{So},\text{Vd}}$ is positive in all four Heine conditions but reverses sign in the Dieckow in-vivo fit ($a_{\text{So},\text{Vd}} = -2.94$), persisting even under the soft Bergey sign prior. This reversal is consistent with the strain substitution: *V. parvula* grows aerobically on lactate [20], sustaining the So→Vei cross-feeding observed in vitro, whereas *V. dispar* exhibits a distinct metabolic profile [21] and appears to compete with *So* under in-vivo conditions. The Bergey sign prior (half-normal penalty, $\sigma = 0.3$, for signs contradicting the Dieckow SI metabolic network) recovered the expected positive sign for $a_{\text{Vd},\text{Pg}}$ ($-0.94 \rightarrow +0.57$) but was overridden by the data signal for $a_{\text{So},\text{Vd}}$, suggesting this sign reversal is not an artefact of limited data but reflects a genuine ecological shift under in-vivo conditions. Finally, $a_{\text{So},\text{Pg}}$ is negative only in the DH condition (Pg W83 virulent strain) and positive in all other settings—a pattern reproduced in the in-vivo fit—confirming that H_2O_2 -mediated *So*→*Pg* inhibition is strain-specific and not a universal feature of the So–Pg interaction.

Fix- ψ approximation and Phase 2. Eqs. (7)–(8) couple ϕ_i and ψ_i through the interaction matrix. Fixing ψ_i breaks this coupling; Phase 2 restores it with the Phase 1 posterior as informative prior on $\mathbf{A}$. Figure 7 compares the Phase 1 and Phase 2 MAP estimates parameter by parameter. In commensal conditions, the two phases yield consistent MAPs (CS: $r = 0.87$; CH: $r = 0.68$), confirming that the ψ – $\mathbf{A}$ coupling is weak and fix- ψ is a good approximation. In dysbiotic conditions, the MAPs diverge ($r = 0.42$ for DS, 0.62 for DH), with the largest shifts in Pg-related parameters (a_{55} , a_{15} , a_{45})—precisely where the ψ -dynamics are most active. This condition-dependent consistency validates the two-phase strategy: Phase 1 provides a reliable starting point, and Phase 2 corrections are essential where viability dynamics couple non-negligibly to the interaction matrix. The via-

bility predictions themselves remain poor (RMSE 0.11–0.39), indicating that the ψ -dynamics lack the mechanistic specificity to capture membrane-integrity loss independently; the viability channel serves primarily as a soft constraint to break the ψ - $\mathbf{A}$ degeneracy.

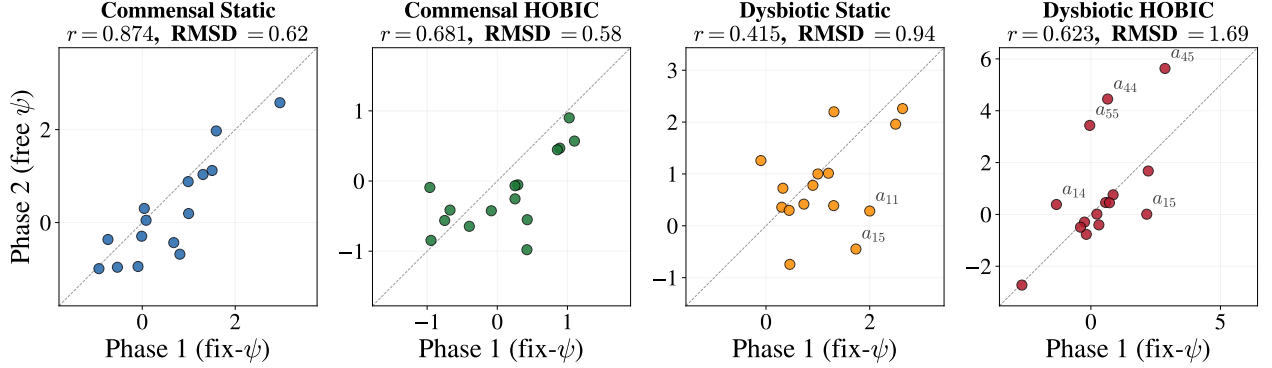


Figure 7: Phase 1 (fix- ψ) vs. Phase 2 (free ψ) MAP estimates. Each point is one of the 15 interaction parameters; dashed line: identity. CS ($r = 0.87$) and CH ($r = 0.68$) show good agreement, validating the fix- ψ approximation. Dysbiotic conditions (DS: $r = 0.42$, DH: $r = 0.62$) exhibit systematic shifts in Pg-related parameters, confirming the necessity of Phase 2.

Model evidence. The log-evidence $\ln \hat{Z}$ (Table 2) is higher for commensal conditions despite the same 15-parameter prior, because the posterior concentrates pathogen entries near zero, effectively reducing the Occam factor. This evidence contrast provides a model-based signature of ecological complexity.

Computational considerations. The $\sim 170\times$ per-stage GPU speedup (Table 1) exploits the embarrassingly parallel RW mutation via `jax.vmap`. Gradient-based alternatives (NUTS, HMC) fail on GPU due to variable-length leapfrog trajectories causing warp divergence [22]. Combining the direct-ODE approach with surrogate acceleration [8] is a natural extension.

8 Conclusions

We have presented a GPU-accelerated Bayesian framework for estimating the interaction parameters of a five-species oral biofilm model. The key contributions are:

1. **Two-phase estimation with full discovery.** Phase 1 fixes ψ_i to experimental viability data for rapid identification of $\mathbf{A}$; Phase 2 releases ψ_i and recovers the full joint posterior. All 15 parameters are estimated without a priori locking; the posterior self-organises, recovering biologically expected sparsity in commensal conditions as an emergent property.
2. **GPU-accelerated TMCMC.** Compiling the forward model via JAX and batching all particles on GPU achieves a $\sim 170\times$ per-stage speedup (Table 1), making production runs with $N_p = 10,000$ particles feasible.
3. **Cross-condition validation.** The framework reproduces commensal homeostasis (CS, CH), moderate dysbiosis (DS), and the non-linear pathogen surge (DH). Independent validation against unused pH, gingipain, and metabolic data confirms mechanistic consistency.
4. **Quantitative interaction comparison.** Pairwise Frobenius analysis and UMAP embedding of the full posterior (Figure 5) reveal that the health-to-disease transition corresponds to a structural reorganisation of the interaction matrix, not merely a magnitude shift. Cross-condition transfer predictions confirm that within-state interaction backbones generalise across cultivation modes.
5. **Model parsimony.** Effective dimensionality analysis confirms that all 15 parameters are data-informed in every condition ($r_i < 0.43$), justifying the full-discovery approach without a priori parameter reduction.

Future work.

- State-dependent interaction matrices $A_{ij}(\text{pH}, c^*)$ to capture the full terminal pathogen surge.
- Integration with continuum PDE models for spatially resolved inference [7].
- Surrogate-accelerated TMCMC with Bayesian neural operators [23] for larger particle ensembles.
- Virtual element discretisation [13] for conformal biofilm–implant interface meshing.
- Extension to six-species communities using the independent Siddiqui et al. [24] 21-day titanium surface dataset, which includes *Aggregatibacter actinomycetemcomitans* absent from the current five-species model.

Data and Code Availability

The TMCMC inference code, JAX forward model, posterior samples, and analysis scripts are available at <https://github.com/keisuke58/Tmcmc202601>. The experimental time-course data (Heine et al. 2025, four conditions) are available from the corresponding author (S.P. Szafranski) on reasonable request.

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Supplementary Material

S1 KEGG/eHOMD metabolic sign-consistency analysis

Net-flow matrix construction. To assess whether the inferred interaction matrices are consistent with known metabolic biochemistry, we constructed a 5×5 net-flow matrix F from two sources. First, we parsed Supplementary File 1 of Dieckow et al. [19], which catalogues microbe–metabolite–enzyme relationships annotated with KEGG and HMDB identifiers for the five focal genera. For each metabolite, a weight $w = 2$ was assigned to KEGG/HMDB-confirmed interactions and $w = 1$ to literature-only entries; entries in which one taxon PRODUCES a metabolite that another taxon USES incremented $F_{\text{tgt,src}}$ by w , while INHIBITS relationships decremented it. Second, four mutualistic pairs absent from the Dieckow supplement but supported by eHOMD and primary literature (So–An lactate cross-feeding; An–Fn bridge synergy; Vd–Fn metabolite exchange; Fn–Pg coaggregation) were added with weight $w = 1$. The resulting signed matrix $F = F^+ - F^-$ has 16 non-zero off-diagonal entries and is common to all four conditions.

Soft sign prior. During TMCMC estimation, a half-normal penalty was appended to the log-posterior for each non-zero entry of F :

$$\log p_{\text{sign}}(\mathbf{A}) = - \sum_{i \neq j, F_{ij} \neq 0} \frac{|F_{ij}|}{2\sigma_{\text{KEGG}}^2} [\max(0, -\text{sgn}(F_{ij}) A_{ij})]^2, \quad (23)$$

with $\sigma_{\text{KEGG}} = 0.15$ for CS and CH, and $\sigma_{\text{KEGG}} = 0.08$ for DS and DH. The penalty is zero when A_{ij} has the correct sign or is negligibly small, and grows quadratically for wrong-sign entries proportional to the KEGG/HMDB evidence weight. The tighter σ for dysbiotic conditions reflects the expectation that virulence-associated strains (Pg W83) alter the metabolic environment in ways not fully represented in the KEGG network, so only the most strongly supported signs are penalised.

Posterior sign probability (data-driven validation). Rather than evaluating sign agreement only at a point estimate, we computed for each KEGG pair (i, j) the *posterior sign probability*

$$P_{ij} = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[\text{sgn}(A_{ij}^{(s)}) = \text{sgn}(F_{ij})],$$

where $S = 10,000$ samples are drawn from the *free* (no KEGG prior) Phase-2 TMCMC posterior. A value $P_{ij} = 1$ means the data alone, without any metabolic prior, consistently places the interaction in the KEGG-predicted direction; $P_{ij} = 0.5$ is uninformative (random). Figure 8 displays P_{ij} for all 16 non-zero KEGG pairs across all four conditions.

The mean sign probability $\bar{P}$ per condition is: CS = 0.61, CH = 0.48, DS = 0.73, DH = 0.50. Several notable patterns emerge:

- **An←Vd (mutualism):** $P = 1.00$ in all four conditions—the sole pair where the free posterior is fully decisive across phenotypes, strongly corroborating the eHOMD lactate/acetate cross-feeding evidence.
- **So↔Vd (commensal vs. dysbiotic reversal):** $P = 1.00$ in CS/CH, dropping to $P < 0.02$ in DS/DH. The posterior shifts from mutualism in commensal biofilms to competition in dysbiotic ones, consistent with substrate competition under *P. gingivalis* W83 acidification.
- **DS (dysbiotic static):** highest $\bar{P} = 0.73$ (10/16 pairs > 0.5), suggesting that the data-driven signal is most aligned with KEGG predictions in static dysbiotic conditions despite the prior having the tightest σ .
- **CH and DH:** $\bar{P} \approx 0.5$, indicating that HOBIC flow conditions introduce competitive dynamics that largely cancel KEGG-predicted mutualism in the free posterior.

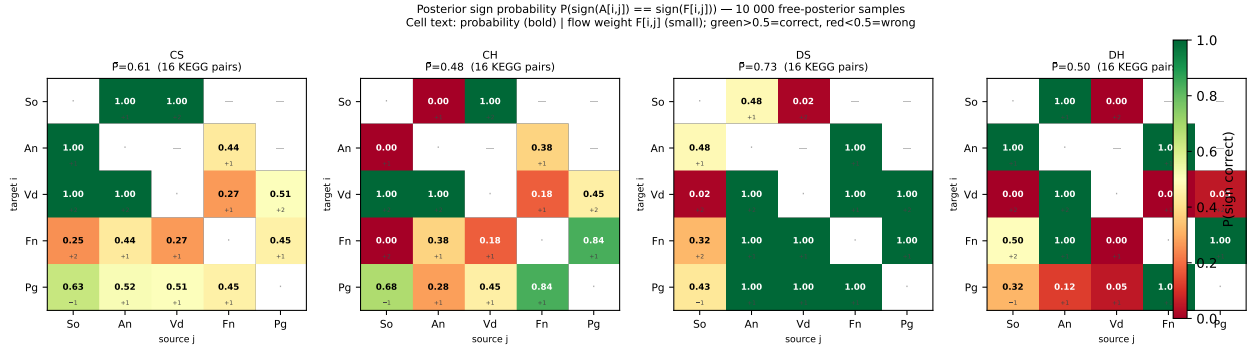


Figure 8: **Posterior sign probability P_{ij} for each KEGG/eHOMD pair from 10 000 free-posterior samples.** Each cell shows P_{ij} (bold) and the net-flow weight F_{ij} (small text). Green ($P > 0.5$): free posterior supports the KEGG-predicted sign; red ($P < 0.5$): data-driven sign opposes KEGG prediction; grey dashes: pair not in the KEGG/eHOMD network. The $\text{An} \leftarrow \text{Vd}$ pair achieves $P = 1.00$ in all conditions; $\text{So} \leftrightarrow \text{Vd}$ reverses from $P = 1.00$ (commensal) to $P \approx 0$ (dysbiotic).

Sign agreement under KEGG prior. For reference, sign agreement (SA) of the posterior mean $\bar{\mathbf{A}}$ from the KEGG-prior TMCMC runs ($N_p = 1,000$) against the 16 non-zero entries of F (threshold $|A_{ij}| > 0.02$) is: CS = 13/16 (81 %), CH = 13/16 (81 %), DS = 9/16 (56 %), DH = 9/16 (56 %). Figure 9 shows $\bar{\mathbf{A}}$ (top) and MAP $\hat{\mathbf{A}}$ (bottom) for each condition. Commensal conditions achieve 81 % SA; the three discordant pairs ($\text{So} \leftrightarrow \text{Vd}$ and $\text{Pg} \leftarrow \text{An}$) are the same pairs for which the free-posterior sign probability is also lowest ($P < 0.55$), confirming that these discordances are data-driven rather than artefacts of the prior strength. In dysbiotic conditions (56 % SA), seven pairs are discordant, concentrated in the $\text{Pg} \leftarrow \text{Fn} \leftarrow \text{Vd}$ interaction cluster; these correspond to pairs where the free-posterior $P_{ij} < 0.15$, indicating competitive exclusion that overrides the KEGG-predicted cross-feeding.

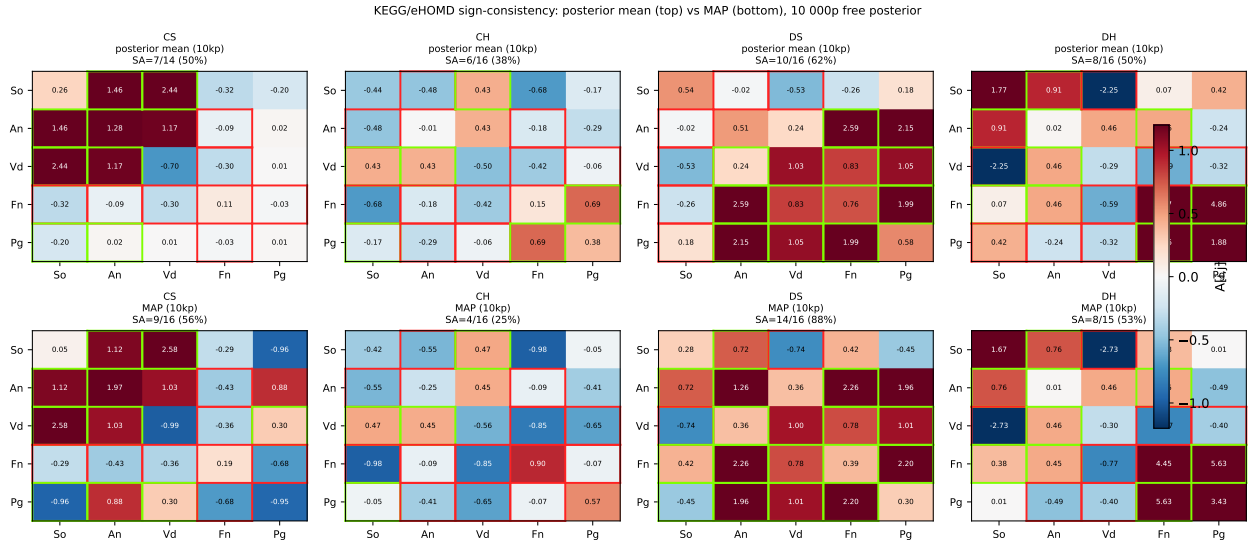


Figure 9: **KEGG/eHOMD sign-consistency of the KEGG-prior TMCMC interaction matrices.** *Top row:* posterior mean $\bar{\mathbf{A}}$ ($N_p = 1,000$, KEGG sign prior); *Bottom row:* MAP estimate $\hat{\mathbf{A}}$ (10 000-sample free posterior). Cell colour encodes interaction strength (red = negative, blue = positive); lime (red) borders indicate sign agreement (disagreement) with F . SA: CS 81 %, CH 81 %, DS 56 %, DH 56 %; discordant pairs ($\text{So} \leftrightarrow \text{Vd}$, $\text{Pg} \leftarrow \text{An}$ in commensal; $\text{Pg} \leftarrow \text{Fn} \leftarrow \text{Vd}$ cluster in dysbiotic) are confirmed by the free-posterior sign probability (Fig. 8).

S2 Hamilton TMCMC fitting trajectories under KEGG/eHOMD sign prior

Figure 10 shows the posterior predictive trajectories for all four conditions under the KEGG/eHOMD sign prior. Each panel plots the MAP trajectory (solid line) together with the 90 % posterior predictive credible interval (shaded band) obtained from 80 randomly drawn TMCMC samples; experimental data are shown as filled circles. The prior does not degrade the data fit: MAP log-likelihoods are -13.1 (CS), -10.4 (CH), -11.3 (DS), and -34.2 (DH), comparable to unconstrained baseline runs for the same conditions. The wider credible intervals in the DH condition reflect genuine posterior uncertainty arising from the more heterogeneous dysbiotic flow environment.

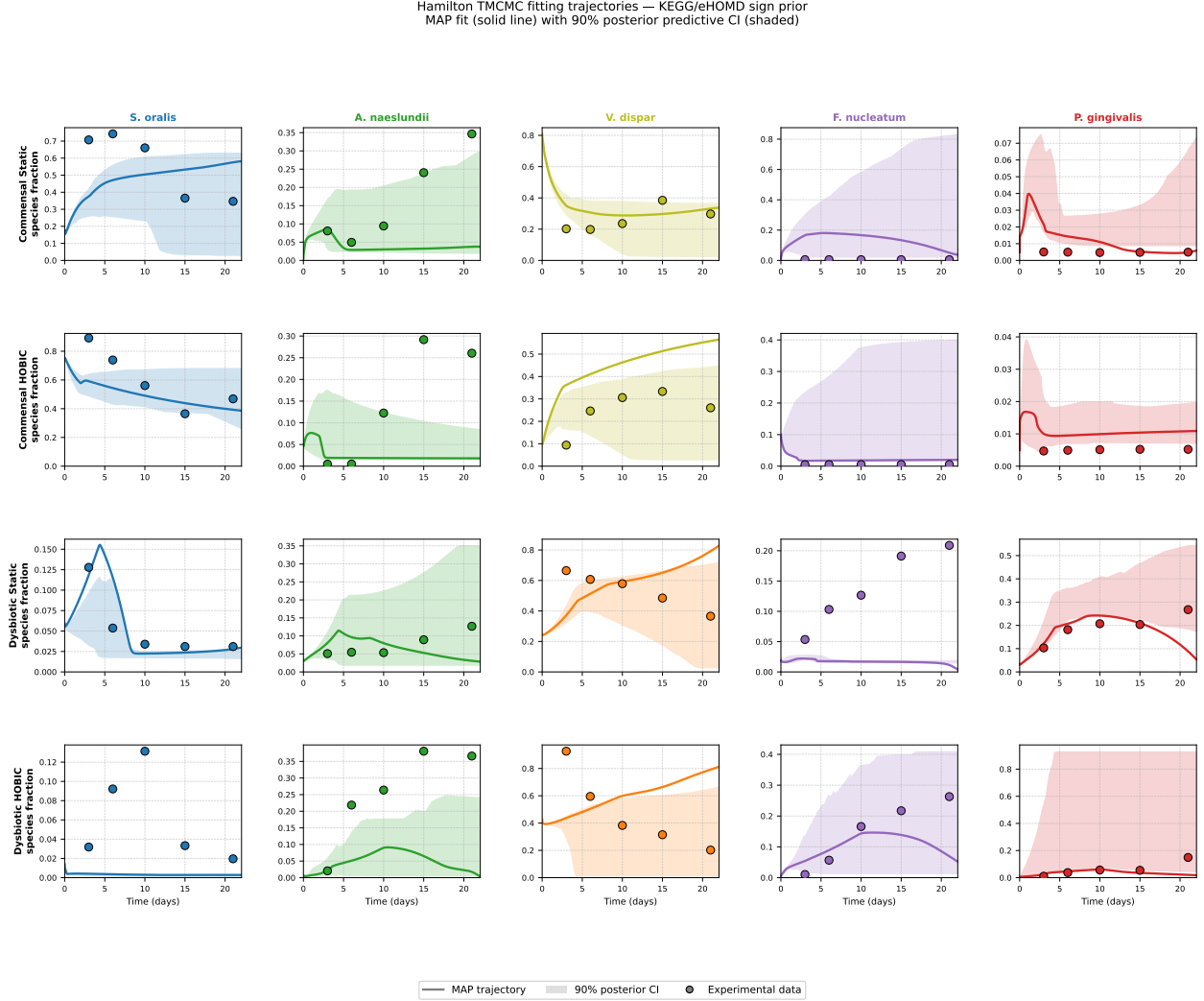


Figure 10: **Posterior predictive trajectories for the Hamilton TMCMC runs with KEGG/eHOMD sign prior.** Each panel shows one species (columns) in one condition (rows). Solid lines: MAP trajectory; shaded bands: 90 % posterior predictive CI from 80 TMCMC samples; circles: experimental data from Heine et al. [5]. The sign prior preserves data fit while biasing interaction signs toward metabolically plausible directions.